

54 Section Diff Ch 4.

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Apr 2026

Ch 4.2

3, 5. Find general solution to the given differential equations

$$y'' + 5y' + 6y = 0, \quad y'' + 8y' + 16y = 0$$

$$r^2 + 5r + 6 = 0$$

$$(r + 3)(r + 2) = 0 \implies r_1 = -3, r_2 = -2$$

Distinct roots implies $\Delta > 0$:

$$c_1 e^{-3t} + c_2 e^{-2t} \quad \square$$

$$r^2 + 8r + 16 = 0$$

$$(r + 4)(r + 4) = 0 \implies r_1 = -4, r_2 = -4$$

Roots are the same $\implies \Delta = 0$:

$$c_1 e^{-4t} + c_2 t e^{-4t} \quad \square$$

15, 17. Solve the initial value problem

$$y'' - 4y' + 3y = 0; \quad y(0) = 1, \quad y'(0) = 1/3$$

$$r^2 - 4r + 3 = 0 \implies (r - 3)(r - 1) \implies y(t) = c_1 e^{3rt} + c_2 e^{rt}$$

$$c_1 + c_2 = 1 \implies c_2 = 1 - c_1$$

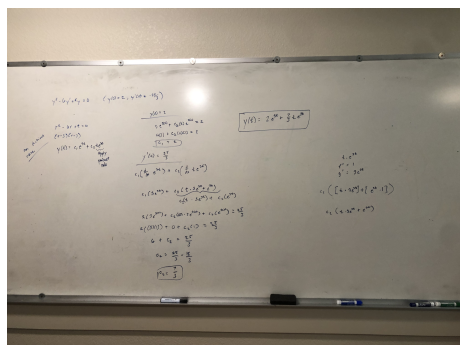
$$3c_1 + c_2 = 1/3 = 3c_1 + (1 - c_1) = 1/3$$

$$= 2c_1 + 1 = 1/3 = 2c_1 = 1/3 - 3/3$$

$$2c_1 = -\frac{2}{3} = c_1 = \frac{2}{1} \times -\frac{3}{2} = -6/2 = -3?$$

In my notes, I solved this correctly with a video. Main idea is that my algebra is rusty. We multiply by the reciprocal of 2/1, not flipping 3/2

$$17. \quad y'' - 6y' + 9y = 0; \quad y(0) = 2, \quad y'(0) = 25/3$$



21. Substituting $y = e^{rt}$, find auxiliary equation for the first-order linear equation $ay' + by = 0$, where a, b are constants with $a \neq 0$. Use the result of the above to find the general solution. **I am able to find auxiliary equation:**

$$ay' + by = 0 \rightarrow a(e^{rt})' + b(e^{rt}) = 0$$

$$a(re^{rt}) + b(e^{rt}) = 0 = e^{rt}(ar + b) = 0 \implies \text{auxiliary equation} = ar + b$$

Then we need to solve for r since this constrains what our general solution will be in the form of. With our asantz equation e^{rt} , we write in the general solution form. Since we only are in one dimension, i.e. y' , then we only have $c_1 e^{rt}$ to find.

$$ar + b = 0$$

$$r + \frac{b}{a} = 0$$

$$r = -b/a \implies c_1 e^{rt} = c_1 e^{[-b/a]t}$$

27-29. Use definition 1 to determine whether the functions y_1, y_2 are linearly independent on the interval $(0, 1)$

Definition

Linear Independence of Two Functions

A pair of functions $y_1(t), y_2(t)$ is L.I. on the interval $I \iff$ neither of them is a constant multiple of the other on all of I . We say that y_1, y_2 are linearly independent on I if one of them is a constant multiple of the other on all of I

27. $y_1(t) = \cos(t) \cdot \sin(t), \quad y_2(t) = \sin(2t)$

- For the given domain that is defined (i.e. $(0, 1)$), our goal is to show the algebraic identity linear independence
- This identity is determined by our division of the two functions, where if it is a constant, then this implies it is dependent.
- Else if t is still present in division, then we are Linearly Independent.

$$W.T.S. \frac{y_1(t)}{y_2(t)} = ct.$$

$$\frac{\cos(t) \cdot \sin(t)}{\sin(2t)} = \frac{\cos(t) \cdot \sin(t)}{\text{Apply sum rule to } y_2}$$

$$y_2 = \sin(2t) = \sin(s+t) = \sin(t+t)$$

$$\implies \sin(s) \cdot \cos(t) + \sin(t) \cdot \cos(s) = \sin t \cdot \cos t + \sin(t) \cdot \cos(t) = 2(\sin(t) \cdot \cos(t))$$

$$\frac{\cos(t) \cdot \sin(t)}{2(\sin(t) \cdot \cos(t))}$$

$$y_1 = \frac{1}{2}y_2 \implies \text{constant} \implies \text{linearly dependent}$$

Therefore the ratio is constant on $(0, 1)$

$$29. y_1(t) = te^{2t}, \quad y_2(t) = e^{2t}$$

$$W.T.S. \frac{y_1(t)}{y_2(t)} = ct \text{ for L.I.}$$

$$\frac{te^{2t}}{e^{2t}} = e^{2t} \frac{t}{1} = e^{2t} \cdot t$$

$$\implies y_1 = e^{2t} \cdot t$$

Thus our ratio is not constant on our domain of $(0, 1)$ evident by our found t , implying it is linearly independent.

$$31. y_1(t) = \tan^2 t - \sec^2 t, \quad y_2(t) \equiv 3$$

$$\frac{\tan^2 t - \sec^2 t}{3} = \text{apply trig identity} \rightarrow \frac{\tan^2 t - (1 + \tan^2 t)}{3}$$

$$\frac{\tan^2 t - 1 - \tan^2 t}{3} = -\frac{1}{3} \implies y_1 = -\frac{1}{3}y_2$$

Constant ratio on our domain $(0, 1)$

- $\equiv$ is nothing more than an indication to not examine its identity too hard and take it for what is is, i.e. in this case 3.

Ch 4.3

Auxiliary equation has complex roots, find general solution.

$$3. z'' - 6z' + 10z = 0$$

Step 1: Auxiliary equation

$$a^2 - 6a + 10 = 0$$

Step 2: Find discriminant

$$(-6)^2 - (10(1)) = -4 \implies \text{complex}$$

Step 3: Find roots

$$\alpha = -\left(\frac{-6}{(2)(1)}\right) = 3$$
$$\beta = \frac{-(-6)\sqrt{(-6)^2 - 4(1)(10)}}{2(1)} = \frac{6\sqrt{-4}}{2} = \frac{6 \times 2i}{2} = \frac{12i}{2} = 6i$$

Step 4: Write in general solution form

$$y(t) = c_1 e^{3t} \cos(6i) + c_2 e^{3t} \sin(6i)$$

Where you got it wrong:

$$\alpha = -\left(\frac{-6}{(2)(1)}\right) = 3$$
$$\beta = \frac{-(-6)\sqrt{(-6)^2 - 4(1)(10)}}{2(1)} = \frac{6\sqrt{-4}}{2}$$
$$r = \frac{6 \pm 2i}{2} = 3 \pm i$$

$$y(t) = c_1 e^{3t} \cos(1(t)) + c_2 e^{3t} \sin(1(t))$$
$$= e^{3t}(c_1 \cos(t) + c_2 \sin(t))$$

- β must always be a real positive number, its the angular frequency.
- β is the imaginary part of the root after full simplification. If i appears inside your trig functions, that is a signal you did not finish dividing.
- $3 \pm i$ has the form $3 \pm 1 \cdot i \implies \beta = 1$. More explicitly, this decomposition can be applied to any constant $c \in \mathbb{R} : 3 \pm c \times i \implies \beta = c$

9. Find general solution of $y'' - 8y' + 7y = 0$

$$r^2 - 8r + 7 = 0 \equiv (r - 7)(r - 1) \rightarrow r_1 = 7, r_2 = 1$$

$\Delta = (8)^2 - 4(1)(7) = 64 - 28 = 36 \implies \Delta = \neg \text{complex}$, write linear combination normally

$$y(t) = c_1 e^{7t} + c_2 e^t$$

- I wrote it as $c_1 e^{7t} + c_2 e^t$ which the textbook writes it as $c_1 e^t + c_2 e^{7t}$. Is there a difference? Why do they put t in the front? Is it because they want the root with 1 to be in the front and its a purely notation logic for the 2 by 2 matrix?

- No difference. c_1 and c_2 are arbitrary constants, so swapping their labels changes nothing. Both expressions span the same solution space.
- The ordering is purely notational convention. Textbooks typically write roots in ascending order, i.e. $r = 1$ before $r = 7$. No linear algebra reason behind it.

17. Find general solution of $y'' - y' + 7y = 0$

21. Solve the initial value problem $y'' + 2y' + 2y = 0$, $y(0) = 2$, $y'(0) = 1$

27. Solve the initial value problem $y''' - 4y'' + 7y' - 6y = 0$; $y(0) = 1$, $y'(0) = 0$, $y''(0) = 0$

$$x = a^3 + b^2 + 7c - 6 = 0 \dots \text{Synthetic_Division}() \dots = a^2 - 2b + 3 = 0 \implies (-2)^2 + 4(1)(3) = 4 - 12 = -8 \implies \Delta = \text{Complex form}$$

Be quadratic form x does not factor over integers, we use quadratic formula to find roots

$$\frac{-(-2) \pm \sqrt{-8}}{2(1)} = \frac{2 \pm \sqrt{-8}}{2} = \frac{2 \pm \sqrt{4 \times -2}}{2} = \frac{2 \pm 2\sqrt{-2}}{2} = 1 \pm \sqrt{\frac{-2}{4}} = 1 \pm -\frac{1}{\sqrt{2}}i$$

$$r_1 = 1 + \frac{1}{\sqrt{2}}, r_2 = 1 - \frac{1}{\sqrt{2}} \perp$$

$$2 \left| \begin{array}{cccc} 1 & -4 & 7 & -6 \\ & 2 & -4 & 6 \\ \hline 1 & -2 & 3 & 0 \end{array} \right| = a^2 - 2b + 3 = 0$$

Where I got it wrong:

1. Division of finding roots and keeping i within the expression

$$\frac{2 \pm \sqrt{-8}}{2} = \frac{2 \pm 2\sqrt{2}i}{2} = 1 \pm \sqrt{2}i \implies r_1 = 1 + \sqrt{2}i, r_2 = 1 - \sqrt{2}i \implies \beta = \sqrt{2} \quad \square$$

35. Swinging Door. The motion of a swinging door with an adjustment screw that controls the amount of friction on the hinges is governed by the IVP

$$I\theta'' + b\theta' + k\theta = 0; \theta(0) = \theta_0, \theta'(0) = v_0$$

where θ is the angle that the door is open, I is the moment of inertia of the door about its hinges, $b > 0$ is a damping constant that varies with the amount of friction on the door, $k > 0$ is the spring constant associated with the swinging door, θ_0 is the initial angle that the door is opened, and v_0 is the initial angular velocity imparted to the door. If I, k , are fixed, determine for which values of b the door will not continually swing back and forth when closing.

- Inertia := the tendency of an object to resist changes in its state of motion

4.4

5-7. Does UC apply?

$$y''(\theta) + 3y'(\theta) - y(\theta) = \sec \theta.$$

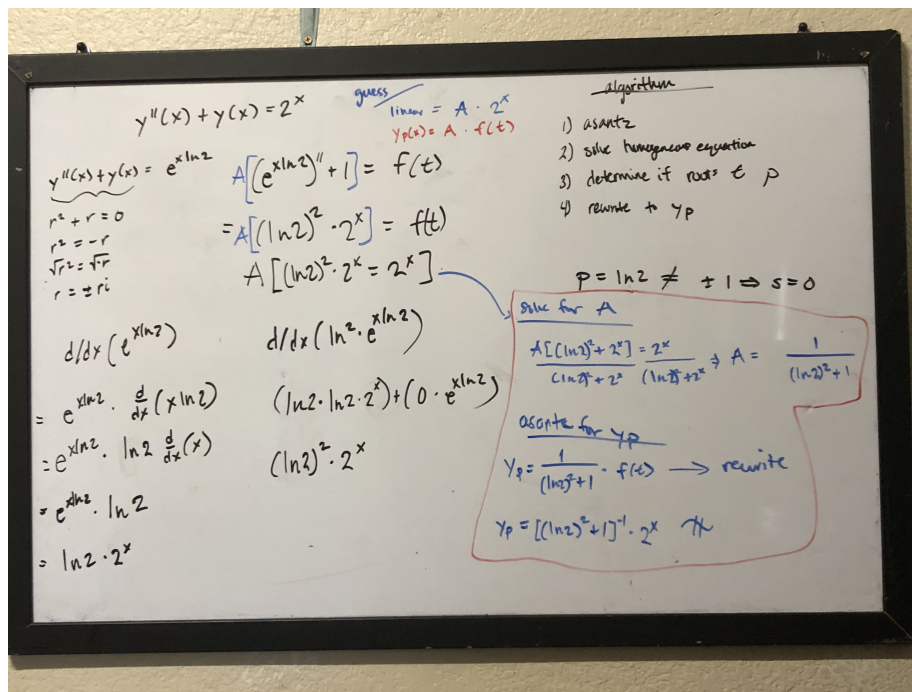
No, no conditions met.

7.

$$5u'(x) - 2u(x) = 7x^{50}e^{-2x}\sin(3x).$$

Yes, because presence of trig, exp, and polynomial

$$11. y''(x) + y(x) = 2^x$$



17. $y'' + 4y = 8 \sin(2t)$ I kept getting stuck on solving for A, B . Lets proceed with the $A \cos(\beta t)$ term:

$$-A \sin(2t) = 2 \sin(2t)$$

$$A \sin(2t) = -2 \sin(2t)$$

$$\frac{A \sin(2t) = -2 \sin(2t)}{(2 \sin(2t))} = A = -2$$

The only time that the above wouldn't pan out correctly, is when you divide the entire expression, i.e. $-A \sin(2t) + B \cos(2t) = 2 \sin(t)$ by $2 \sin(t)$, because then we would be left with $-A + B \cot(2t) = 2$. Now B is stuck with a function of t .

Solution

- When we solve for A , here is the intuition behind it.
- $\sin(2x), \cos(2x)$ are orthogonal functions, meaning they are independent. You can't express one as a multiple of the other.
- Thus, if $A \sin(2x) + B \cos(2x) = C \sin(2x) + D \cos(2x) \forall x$, then $A = C$ and $B = D$, separately I.e. you cannot have $A \neq C$ and have the equation hold $\forall x$

23. $y''(\theta) - 7y'(\theta) = \theta^2$

- You must recognize that we have a polynomial, thus we have the case of $t^m(A_mt^m + \dots + A_1t^1 + A_0t^0)$
- Note: We use a higher degree because there is no constant term in the original equation)
- Since these must be equal for all θ , you get the forced system:

$$\begin{aligned} -21A &= 1 \\ 6A - 14B &= 0 \\ 2B - 7C &= 0 \end{aligned}$$

- This is a triangular linear system — solve top-down sequentially A first, substitute into the second equation for B , then substitute B into the third for C . It is the only systematic method for this form.
- To build this intuition, you need to recognize this as a linear system $Ax = b$ where the matrix is triangular by construction, making back-substitution the natural solver.
- If the system were not triangular, you'd need full Gaussian elimination. Here the structure is guaranteed by degree-matching order.

27. $y'' + 9y = 4t^3 \sin(3t)$

- for sake of time, I need to move on. I almost got the set up right. For future me, observe:

$r^2(e^{rt}) + 9(e^{rt}) = 0 \implies r = \pm 3i$ meaning we need to have a t so we don't solve for homogeneous equation
Asantz RHS :

$$e^{\alpha t} \times 4(t[(At^4 + At^3 + At^2 + At)\cos(3t) + (Bt^4 + Bt^3 + Bt^2 + Bt)\sin(3t)])$$

4.5

1. Given that $y_1(t) = \cos(t)$ is a solution to $y'' - y' + y = \sin(t)$, and $y_2(t) = e^{2t}/3$ is a solution to $y'' - y' + y = e^{2t}$.

Use the superposition principle to find solutions to the following differential equations:

$$\begin{aligned} y'' - y' + y &= 5 \sin(t) \\ ey'' - y' + y &= \sin(t) - 3e^{2t} \\ y'' - y' + y &= 4\sin(t) + 18e^{2t} \end{aligned}$$

Steps:

- Identify roots of auxiliary equation

- Check if roots from aux eq appears in RHS forcing function, if so, apply t . Not, do nothing
- Check multiplicity of $t \in f(t)$
- Plug $asantz \in y'' - y' + y = f(t)$. Solve in terms of A . This is the first part of the equation. Repeat the above to find the superposition equation, y .

Identify roots of auxiliary equation:

$$aux = r^2 - r + 1 = 0 \rightarrow quadraticformula(aux) = \frac{1}{2} \pm \frac{\sqrt{3}}{2}i$$

$$s = 0, m = 0, \alpha = \frac{1}{2}, \beta = \frac{\sqrt{3}}{2}$$

Plug $asantz \in y'' - y' + y = f(t)$. Solve in terms of A .

$$(A\cos(t) + B\sin(t))'' + (A\cos(t) + B\sin(t))' + (A\cos(t) + B\sin(t)) = \sin(t)$$

$$= [-A\cos(t) - B\sin(t)] - [-A\sin(t) + B\cos(t)] + [A\cos(t) + B\sin(t)] = \sin(t)$$

$$= A\sin(t) - B\cos(t) = \sin(t)$$

$$(1 \cdot \sin(t)) + (0 \times \cos(t)) \implies \text{We only solve for } A \sin(t)$$

$$A\sin(t) = \sin(t)$$

$$\frac{A\sin(t)}{\sin(t)} = \frac{\sin(t)}{\sin(t)} = A = 1$$

Thoughts:

- Even though I found $A = 1$, this was an incorrect process. I was supposed to just do coefficient matching here and identify that we have $(1 \cdot \sin(t)) + (0 \times \cos(t))$, where we see $1 = A$.
- For $y'' - y' + y = 5\sin(t)$, you correctly found the LHS. You just need to do:

$$y'' - y' + y = 5\sin(t)$$

Plug $asantz \in y'' - y' + y = f(t)$ to get:

$$A\sin(t) - B\cos(t) = 5\sin(t)$$

$$\implies A = 5 \text{ ??? still valid}$$

✓ Simple coefficient matching:

$$[A \times \cos(t)] + [B \times \sin(t)] = [5 \times \cos(t)] + [0 \times \sin(t)]$$

$$\implies y_p = 5\cos(t)$$

- I originally tried to do $A\sin(t) - B\cos(t) = \cos(t)$ which is a contradiction. A should be a constant. NOT $\cot(t)$ which is a function (when you solve that's what you get)... this signals an upstream error (wrong RHS target). Pure absurdity on my part, keep it equal to the forcing function $\sin(t)$.
- Solving via undetermined coefficients was the incorrect path here.
- Rather, since are given $y_1 = \cos(t)$ as particular solution, you would do the following:

$$y'' - y' + y = 5\sin(t) \rightarrow y_p = 5y_1(t) = 5\cos(t) \quad \square$$

Leading with our above intuition, let's just do coefficient matching.

We know: $y'' - y' + y = \sin(t)$ gives $y_1(t) = \cos(t) \wedge y'' - y' + y = e^{2t}$ gives $y_2(t) = \frac{e^{2t}}{3}$. So it follows:

$$\begin{aligned} y'' - y' + y &= \sin(t) - 3e^{2t} = y_1(t) - 3(y_2(t)) = \\ &= y_1(t) - 3\frac{e^{2t}}{3} = \cos(t) - e^{2t} \quad \square \end{aligned}$$

$$\begin{aligned} y'' - y' + y &= 4\sin(t) + 18e^{2t} = 4(y_1(t)) + 18(y_2(t)) \\ &= 4\cos(t) + \left[\frac{18}{1} \cdot \frac{e^{2t}}{3}\right] = 4\cos(t) + 6e^{2t} \quad \square \end{aligned}$$

7. A nonhomogeneous equation and a particular solution are given. Find a general solution for the equation.

$$y'' = 2y + 2\tan^3 x, \quad y_p(x) = \tan(x)$$

Proposed Algorithm:

1. We are given y_p . Let's find y_h
2. We find y_h by setting the forcing function in our equation to be 0, THEN setting equal to 0.
3. After step 2, we will set up the auxiliary equation to find our roots, i.e. finding $y_{complementary}$, then we plug in the found roots $\{\alpha_1, \alpha_2\}$ into $c_1 e^{\alpha_1 x} + c_2 e^{\alpha_2 x}$.
4. Now just write it in the final form $y = y_h + y_p$

$$\begin{aligned} y_h &= y'' = 2y + 2\tan^3 x \\ y_h &= y'' = 2y + 2(0) \\ y_h &= y'' - 2y = 0 \rightarrow r^2 - 2 = 0 = r = \pm\sqrt{2} \\ y_c &= c_1 e^{\sqrt{2}x} + c_2 e^{-\sqrt{2}x} \end{aligned}$$

$$y = c_1 e^{\sqrt{2}x} + c_2 e^{-\sqrt{2}x} + \tan(x)$$

- The logic is $\mathcal{L}[y_h + y_p] = \mathcal{L}[y_h] + \mathcal{L}[y_p] = 0 + f(x) = f(x)$
- y_h must satisfy $\mathcal{L}[y_h] = 0$. That's why we set $f(x) = 0$
- $c_1 e^{\alpha_1 x} + c_2 e^{\alpha_2 x}$ is not a guess. It is the **kernel of $\mathcal{L}$** spanned by linearly independent solutions. The constants in the equation account for all initial conditions!!

9. Decide whether the method of undetermined coefficients together with superposition can be applied to find a particular solution of the given equation. Do not solve the equation

$$3y'' + 2y' + 8y = t^2 + 4t - t^2 e^t \sin(t)$$

I say true because it has forcing functions that have asantz's we have seen before. I.e. $t^2, e^t, \sin(t)$. Our LHS is the most general linear case as well.

True

- The reason why they ask you not to solve it is because it is an involved equation. Maybe try this for extra practice.

21. Find a general solution to the differential equation $y''(\theta) + 2y'(\theta) + 2y(\theta) = e^{-\theta} \cos(\theta)$

Initial thoughts on this problem:

1. I solve the auxiliary equation
2. I write the roots found into the asantz equation of the forcing function
1. $r^2 + 2r + 2 = 0 \implies \text{quadratic formula}() \rightarrow \text{roots} = -1 \pm i, m = 0, s = 1$
2. $Ae^{(-1)(t)} \cos((1)(t)) + Bte^{(-1)(t)} \sin((1)(t))$

I correctly identified that $s = 1$, however, when you apply the extra t , you must apply it to the entire asantz by t^s . A, B are two terms that form the trial solution, they are not independent. Splitting t^s between them breaks the structure of the ansatz... so $y_p = t^1(Ae^{-t} \cos(t) + Be^{-t} \sin(t))$

Correct process:

9.1

Express the given system of differential equations in matrix notation

1.
$$\begin{aligned} x' &= 7x + 2y \\ y' &= 3x - 2y \end{aligned}$$
3.
$$\begin{aligned} x' &= x + y + z \\ y' &= 2z - x \\ z' &= 4y \end{aligned}$$

$$5. \quad \begin{aligned} x' &= (\sin t)x + e^t y \\ y' &= (\cos t)x + (a + bt^3)y \end{aligned}$$

Very simple questions. Demonstrating what to do with 5 since it looks more scary

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} (\sin t) & e^t \\ (\cos t) & (a + bt^3) \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

9. Express the given higher order differential equation as a matrix system in normal form:
Airy equation $y'' - ty = 0$

- The logic is: reduce order by defining state variables, then express each derivative in terms of the state vector.
- For some reason, this simple problem took me a long time to understand. I feel that I still don't understand it 100%. For example, why would we solve for x'_2 and not x'_1 ?

Textbook Solution: Let $x_1 = y$, $x_2 = y'$, then $x'_1 = y'$, $x'_2 = y''$

$$\begin{bmatrix} x'_1 \\ x'_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ t & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

For some reason the notation is messing me up so I decided to proceed with this shown by my really smart persian friend. See:

$$\begin{bmatrix} y \\ y' \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} y' \\ y'' \end{bmatrix} \rightarrow \begin{aligned} ay + by' &= y' \\ cy + dy' &= y'' \end{aligned}$$

What does a, b, c, d have to be for it to equal the forcing function on RHS?

$$\begin{aligned} (0)y + (1)y' &= y' \\ cy + dy' &= y'' \end{aligned}$$

$$\text{notice : } y'' = ty \implies cy + dy' = y'' = ty \implies cy + dy' = ty$$

$$(t)y + (0)y' = ty \implies \begin{bmatrix} 0 & 1 \\ t & 0 \end{bmatrix}$$

11. Express the given system of higher order differential equations as a matrix system in normal form.

$$\begin{aligned} x'' + 3x + 2y &= 0 \\ y'' - 2x &= 0 \end{aligned}$$

- Guide to dealing with these types of questions

- [Homework examples](#)

IMO the textbook is really ass at describing the computational methods. Or maybe its finally clicking. I recommend visiting the links I attached above. Nonetheless, here is a shitty algorithm:

- Helper functions should write the variable in terms of the first system of equations. I.e., use x instead of y , since our first row starts with $x'' \in x'' + 3x + 2y = 0$. In other words, top down approach for constructing A .
- I noticed that in these problems we usually have our first row acting as $[0 \ 1 \ (n-1)]$. I.e. when we take the derivative of x_1 , we get $x'_1 = x_2$.
- When we take the derivative of x'_2 for example, we usually get what is in our system of equations. Because we have that identity, that allows us to write $x'_2 = - \text{something rearranged}$ in terms of its highest derivative within the system of equations. Please see below example for intuition.

$$\text{Let } x = \begin{Bmatrix} x_1 = x \\ x_2 = x' \\ x_3 = y \\ x_4 = y' \end{Bmatrix}, x' = \begin{Bmatrix} x'_1 = x_2 \\ x'_2 = x'' \implies (x'' = -3x_1 - 2x_3) \\ x'_3 = x_4 \\ x'_4 = y'' \implies (y'' = 2x_1) \end{Bmatrix}$$

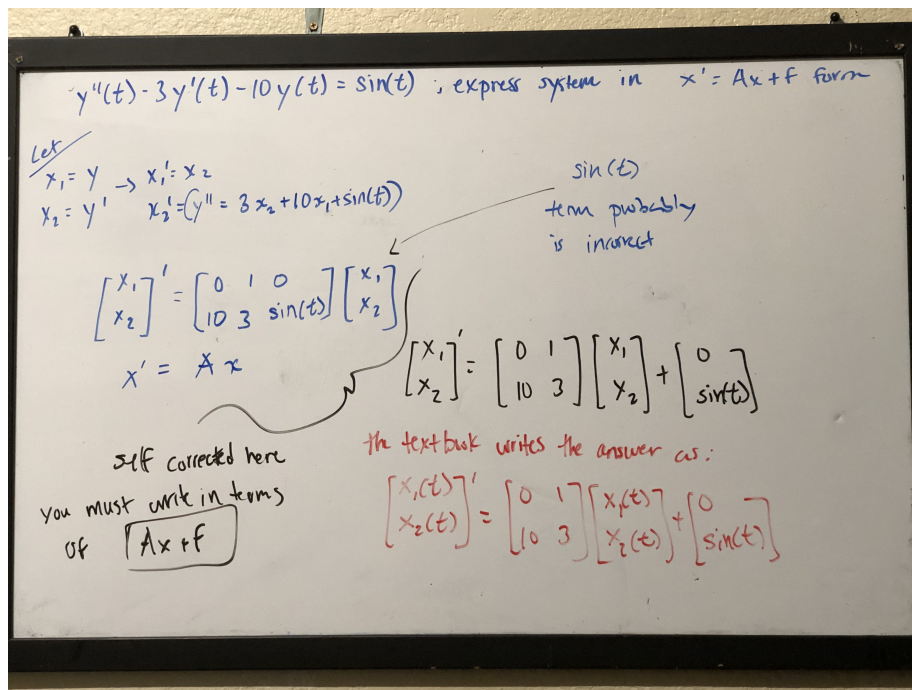
Convince yourself of this (i.e. do it on your own), then construct the normal form:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}' = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -3 & 0 & -2 & 0 \\ 0 & 0 & 0 & 1 \\ 2 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \quad \square$$

9.4

5. Rewrite the given scalar equations as a first order system in normal form. Express the system in the matrix form $\mathbf{x}' = \mathbf{A}\mathbf{x} + \mathbf{f}$.

$$y''(t) - 3y'(t) - 10y(t) = \sin t$$



9. Write the given system as a set of scalar equations

$$\mathbf{x}' = \begin{bmatrix} 5 & 0 \\ -2 & 4 \end{bmatrix} \mathbf{x} + e^{-2t} \begin{bmatrix} 2 \\ -3 \end{bmatrix}$$

$$x_1' = 5x_1 + 2e^{-2t}$$

$$x_2' = -2x_1 + 4x_2 - 3e^{-2t}$$

- Simple problem, just don't forget to apply the t parameter β . Simply multiply and set up the equation. Translated from matrix form into scalar equations.
- Don't get tripped up on the last vector for the $f(t)$. It is linear w.r.t its system of equations, i.e. x_1' has $2e^{-2t}$ and not $-3e^{-2t}$ in it

19. Determine whether the given vector functions are linearly dependent or linearly independent on the interval $(-\infty, \infty)$

$$\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} t \\ 0 \\ t \end{bmatrix}, \begin{bmatrix} t^2 \\ 0 \\ t^2 \end{bmatrix}$$

- Using Wronskian det to find independence conditions

Textbook algorithm:

- Linear independence implies that $\forall t \in (\text{given domain}), c_1, c_2, \dots, c_n = 0$

- We can force $t = 0$ to allow us to just use the coefficients of $\mathbf{x}' = \mathbf{Ax} + \mathbf{f}$ in vector form. Then we can solve it, or compute the determinant.
- I solve for it with algebra, then I use the $\det(A) \neq 0$ lemma that means we have L.I.
- In both implementations below, we set $t = 0$ then solve, and we say it holds for all cases.

Solving through algebra:

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix} = \begin{matrix} c_1 + c_2 + c_3 \\ 0 + 0 + 0 \\ c_1 + c_2 + c_3 \end{matrix}$$

$$\begin{matrix} c_1 = -c_2 - c_3 \\ c_2 = 0 \\ c_3 = -c_1 - c_2 \end{matrix} \rightarrow \begin{matrix} c_1 = -c_3 \\ c_2 = 0 \\ c_3 = -(-c_3) = c_3 \end{matrix} \Rightarrow \text{free var by definition} \Rightarrow L.D.$$

Solving through determinants (put in upper triangular):

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \det() = 0 \Rightarrow L.D.$$

21. The given vector functions are solutions to a system $\mathbf{x}'(t) = \mathbf{Ax}(t)$. Determine whether they form a fundamental solution set. If they do, find a fundamental matrix for the system and give a general solution.

$$\mathbf{x}_1 = e^{2t} \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \quad \mathbf{x}_2 = e^{2t} \begin{bmatrix} -2 \\ 4 \end{bmatrix}$$

Proposed algorithm:

- Simply set up the wronskian $W(t) = [\mathbf{x}_1 \quad \mathbf{x}_2]$
- If $\det(W(t)) = 0 \Rightarrow L.D.$. So put in upper triangular form to take the diagonal OR just take the determinant. For learning purposes I did both. Simple ideas that carry over here.

$$W(t) = \begin{bmatrix} e^{2t} \\ -2e^{2t} \end{bmatrix} \begin{bmatrix} -2e^{2t} \\ 4e^{2t} \end{bmatrix} \rightarrow \det(W(t)) = (e^{2t})(4e^{2t}) - (-2e^{2t})(-2e^{2t}) = 4e^{4t} - 4e^{4t} = 0 \Rightarrow \text{Not fundamental}$$

Upper triangular way:

$$W(t) \rightarrow \begin{bmatrix} e^{2t} & -2e^{2t} \\ -2e^{2t} & 4e^{2t} \end{bmatrix} \sim \begin{bmatrix} e^{2t} & -2e^{2t} \\ 0 & 0 \end{bmatrix} \Rightarrow \det = 0, \text{ Can simply just stop here too since we clearly see L.D.}$$

9.5

15. Find a general solution of the system $\mathbf{x}'(t) = \mathbf{A}\mathbf{x}(t)$ for the given matrix $\mathbf{A}$

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 2 & 1 & 2 \end{bmatrix}$$

Proposed Algorithm:

- Find the roots (characteristic equation), then respective eigenvalues and eigenvectors
- If the matrix can be satisfied by a $n - 1$ scalar equation out of n columns or rows, then represent the "free" one as s so we can claim its linearly independent, where we set $\mathbf{u}_{(n-1)} = s \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
- If linearly independent, then with eigenvalues, write the general solution as: $\mathbf{x}'(t) = c_1 e^{r_1} \mathbf{E}_{\lambda_1} + c_2 e^{r_2} \mathbf{E}_{\lambda_2}$

Handwritten solution for problem 15:

Characteristic equation: $\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} 1-\lambda & 2 & 3 \\ 0 & 1-\lambda & 0 \\ 2 & 1 & 2-\lambda \end{vmatrix} = 0$

Expanding the determinant: $(1-\lambda)(2-\lambda)(2-\lambda) - 6 = 0$

Simplifying: $(1-\lambda)(2-\lambda)^2 - 6 = 0$

Expanding: $(1-\lambda)(4-4\lambda+\lambda^2) - 6 = 0$

Simplifying: $4-4\lambda+\lambda^2-4\lambda+4\lambda^2-\lambda^3-6 = 0$

Combining like terms: $-\lambda^3+5\lambda^2-8\lambda-2 = 0$

Factoring: $(\lambda-2)(\lambda^2-3\lambda-1) = 0$

Roots: $\lambda_1 = 2, \lambda_2 = \frac{3+\sqrt{13}}{2}, \lambda_3 = \frac{3-\sqrt{13}}{2}$

Eigenvectors:

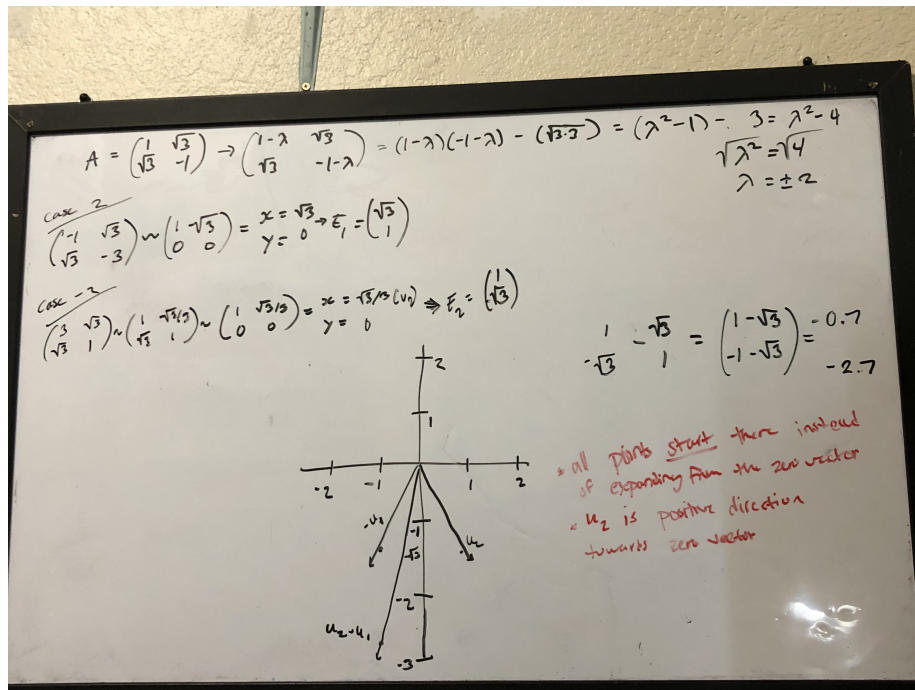
- For $\lambda_1 = 2$: $\begin{bmatrix} -1 & 2 & 3 \\ 0 & -1 & 0 \\ 2 & 1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & -3 \\ 0 & -1 & 0 \\ 0 & 5 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 0 \\ 0 & 0 & 6 \end{bmatrix}$. Let $x = z$, $y = 0$, $z = 0$. Eigenvector: $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$
- For $\lambda_2 = \frac{3+\sqrt{13}}{2}$: $\begin{bmatrix} 1-\lambda_2 & 2 & 3 \\ 0 & 1-\lambda_2 & 0 \\ 2 & 1 & 2-\lambda_2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1/4 \\ 0 & 1 & 3/2 \\ 0 & 0 & 0 \end{bmatrix}$. Let $x = 1/4$, $y = -3/2$, $z = 0$. Eigenvector: $\begin{bmatrix} 1/4 \\ -3/2 \\ 0 \end{bmatrix}$
- For $\lambda_3 = \frac{3-\sqrt{13}}{2}$: $\begin{bmatrix} 1-\lambda_3 & 2 & 3 \\ 0 & 1-\lambda_3 & 0 \\ 2 & 1 & 2-\lambda_3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1/4 \\ 0 & 1 & 3/2 \\ 0 & 0 & 0 \end{bmatrix}$. Let $x = -1/4$, $y = 3/2$, $z = 0$. Eigenvector: $\begin{bmatrix} -1/4 \\ 3/2 \\ 0 \end{bmatrix}$

General solution: $\mathbf{x}(t) = c_1 e^{2t} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + c_2 e^{\frac{3+\sqrt{13}}{2}t} \begin{bmatrix} 1/4 \\ -3/2 \\ 0 \end{bmatrix} + c_3 e^{\frac{3-\sqrt{13}}{2}t} \begin{bmatrix} -1/4 \\ 3/2 \\ 0 \end{bmatrix}$

17. Consider the system $\mathbf{x}'(t) = \mathbf{A}\mathbf{x}(t) \geq 0$ with $\mathbf{A} = \begin{bmatrix} 1 & \sqrt{3} \\ \sqrt{3} & -1 \end{bmatrix}$

1. Show that the matrix $\mathbf{A}$ has eigenvalues $r_1 = 2, r_2 = -2$ with corresponding eigenvectors $\mathbf{u}_1 = \text{col}(\sqrt{3}, 1), \mathbf{u}_2 = \text{col}(1, -\sqrt{3})$

2. Sketch the trajectory of the solution having initial vector $\mathbf{x}(0) = -\mathbf{u}_1$
3. Sketch the trajectory of the solution having initial vector $\mathbf{x}(0) = \mathbf{u}_2$
4. Sketch the trajectory of the solution having initial vector $\mathbf{x}(0) = \mathbf{u}_2 - \mathbf{u}_1$



17. (b)

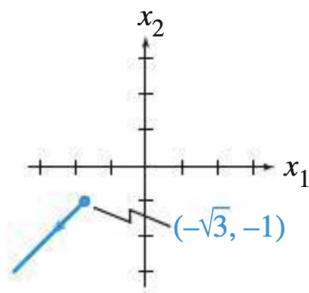


Figure B.43

(c)

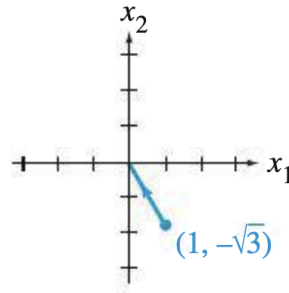


Figure B.44

(d)

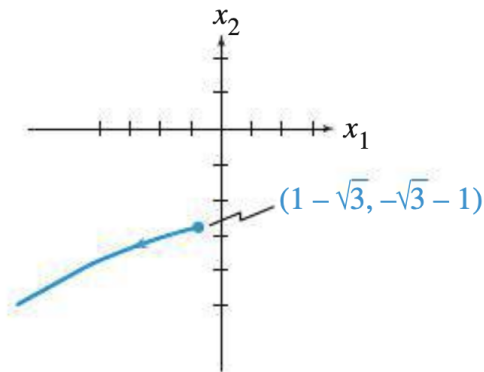


Figure B.45

The general solution we found was $\mathbf{x}'(t) = c_1 e^{2t} \begin{pmatrix} \sqrt{3} \\ 1 \end{pmatrix} + c_2 e^{-2t} \begin{pmatrix} 1 \\ -\sqrt{3} \end{pmatrix}$.

- Say we decided to input 2 or any constant instead of 0 for learning purposes, then e^2, e^{-2} would scale each eigenvector component respectively — it is not a uniform scalar, it is differential scaling along each eigen direction.
- In a picture sense, **the trajectory moves along the curve, not just rescales the point uniformly.**
- That's why in the photos of the graph you see it move with the curve even at 0

19. Find a fundamental matrix for the system $\mathbf{x}'(t) = \mathbf{A}\mathbf{x}(t)$ for the given matrix $\mathbf{A}$

$$\mathbf{A} = \begin{bmatrix} -1 & 1 \\ 8 & 1 \end{bmatrix}$$

Algo:

- We find the eigenvalues, then we write the fundamental equation as $[u_1 \ u_2]$ where $u_1 = e^{\lambda_1 t} \cdot E_{\lambda_1}, u_2 = e^{\lambda_2 t} \cdot E_{\lambda_2}$
- Why? **They represent the corresponding solution vectors for our ODE system, making up the fundamental matrix system**
- Great source
- DO not mix up the general solution and the fundamental equation. Defined as:
 - A fundamental matrix is a matrix whose columns are linearly independent solutions $\Psi(t)$.
 - While the general solution is the vector $x(t) = \Psi(t)c$ (where c is the constant vector) that represents all possible solutions to a linear system

$$\begin{pmatrix} -1 & 1 \\ 8 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} -1-\lambda & 1 \\ 8 & 1-\lambda \end{pmatrix} = (-1-\lambda)(1-\lambda) - (8)(1) = \lambda^2 - 1 - 8 = \lambda^2 - 9 \Rightarrow \sqrt{\lambda^2} = \sqrt{9}$$

$$\lambda = \pm 3$$

$$\Psi(t) = e^{\lambda_1 t} u_1 + e^{\lambda_2 t} u_2$$

$$\frac{1}{4} \cdot 4 = \frac{4}{4} = 1$$

case 3

$$\begin{pmatrix} -2 & 1 \\ 8 & -2 \end{pmatrix} \sim \begin{pmatrix} 1 & -1/4 \\ 8 & -2 \end{pmatrix} \sim \begin{pmatrix} 1 & -1/4 \\ 0 & 0 \end{pmatrix} = \begin{matrix} x = 1/4 \\ y = 0 \end{matrix} \Rightarrow \begin{pmatrix} 1 \\ 4 \end{pmatrix} = E_1$$

case -3

$$\begin{pmatrix} 2 & 1 \\ 8 & 4 \end{pmatrix} \sim \begin{pmatrix} 1 & 1/2 \\ 8 & 4 \end{pmatrix} \sim \begin{pmatrix} 1 & 1/2 \\ 0 & 0 \end{pmatrix} = \begin{matrix} x = 1/2 \\ y = 0 \end{matrix} \Rightarrow E_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\frac{1}{2} \cdot 2 = \frac{2}{2} = 1$$

fundamental solution

$$e^{3t} \begin{pmatrix} 1 \\ 4 \end{pmatrix} + e^{-3t} \begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{bmatrix} e^{3t} & e^{-3t} \\ 4e^{3t} & -2e^{-3t} \end{bmatrix}$$

25. Using matrix algebra techniques, find a general solution of the system:

$$\begin{aligned} x' &= x + 2y - z \\ y' &= x + z \\ z' &= 4x - 4y + 5z \end{aligned}$$

Algo:

- This is already in first order form. We do not have to do the steps from ch 9.4 (i.e. let $x_1 = x, x_2 = x'$. We can simply just take the coefficients from the system of equations and set up our matrix:

$$\begin{bmatrix} 1 & 2 & -1 \\ 1 & 0 & 1 \\ 4 & -4 & 5 \end{bmatrix}$$

- Computationally, I made a lot of mistakes on this problem. When you subtract $0 - \lambda$, you get its eigen value to be $-\lambda$ in the middle row.
- Through cofactor expansion, you can show that the three terms have $(1 - \lambda)$ in common, and from there you can identify the roots. I think thats why they wanted us to do this problem. I usually just to synthetic division.
- Once you find the general solution and write it in $\sum_{i=1} c_i e^{\lambda_i t} E_{\lambda_i}$, make sure to set it up as:

x = First linear combination of Row 1

y = Second linear combination of Row 2

nth = Last linear combination of Row n

33. Solve the given initial value problem:

$$\mathbf{x}'(t) = \begin{bmatrix} 1 & -2 & 2 \\ -2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix} \mathbf{x}(t), \quad \mathbf{x}(0) = \begin{bmatrix} -2 \\ -3 \\ 2 \end{bmatrix}$$

Algo:

1. Find eigenvalues
2. Find eigenvector
3. Write in general solution with found eigenvalues $e^{(roots)} \cdot E_n$
4. Set the eigenvalues found in step 3 equal to $\mathbf{x}(0)$, then reduce the row in the upper triangular form, and back substitute. This is your answer.

Side note, this problem is really computationally annoying, there is a trick where you pull out the common term $(\lambda + 1)$ in this case to find the roots. But the idea is important because this computation lets us do $o(n)$ computation time instead of $o(n^3)$ when computing ML and Markov programs, more as well. Just mentioned the ones that I find interesting.

9.6

3. Find a general solution of the system $\mathbf{x}(t)' = \mathbf{A}\mathbf{x}(t)$ for the given matrix.

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix}$$

- Same steps as in Section 9.5, problem 33
- Note, that calculators like to un-negate the negate power in the equations. I.e. they use $-(\lambda - 1)$, which is equivalent to $-\lambda + 1$
- This problem shows you that RREF does NOT change null space of $(A - \lambda I)$, meaning eigenvector is preserved
- The i persisting in the eigenvector is not a bug, it is correct
- Something you frequently misremember, is that if we have a complex conjugate, i.e. in this problem we get $1 \pm i$, then the second pair $(1 - i)$ provides the same information about the system's growth rate, oscillation frequency, and stability, just with opposite orientation. So therefore its redundant to find its eigenvector, because its the same, i.e $a \pm \beta \equiv a - \beta$ in eigen decomposition.
- Recall that the proof behind that is $Av = \lambda v \equiv A\bar{v} = \bar{\lambda}\bar{v}$, assuming A is real then conjugating the real will yield $\bar{A} = A$.
- We tend to write the real part of the equation after the complex part when we write the general solution.

Complex Conjugate: $-\lambda^3 + 3\lambda^2 - 4\lambda + 2 \rightarrow \text{synthetic_division}() = \lambda_1 = 1$
 $-\lambda^2 + 2\lambda - 2 \rightarrow \text{quadratic_formula}() = \lambda_{2,3} = 1 + i$

Case $\det(A - 1)$:

$$\begin{bmatrix} 0 & 2 & -1 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix} \rightarrow \text{rref}(A) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = E_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Case $\det(A - (1 + i))$:

$$\begin{bmatrix} -i & 2 & -1 \\ 0 & -i & 1 \\ 0 & -1 & -i \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -ix_1 + 2x_2 - x_3 \\ -ix_2 + x_3 \\ -x_2 - ix_3 \end{bmatrix}$$

$$-ix_2 + x_3 = 0 \rightarrow x_3 = -ix_2$$

$$\begin{aligned} x_3 = ix_2 = i(1) = i &\implies x_3 \in \text{Row 3} = -x_2 - ix_3 = -(1) - i(i) \\ &= -1 - i^2 = -1 - -(1) = -1 + 1 = 0 \\ &\implies R3 \text{ redundant} \end{aligned}$$

We say x_2 has to be one because its a pivot and:

$$ix_2 = -x_3 = ix_2 = -i \rightarrow x_2 = -i / -i \implies x_2 = 1$$

$$\begin{aligned} -ix_1 + 2x_2 - x_3 &= -ix_1 + 2(1) - i = x_1 = \frac{-2 + i}{-i} \\ x_1 &= -1 - 2i \end{aligned}$$

After being lost in the algebra, we set up the eigenvector this case:

$$\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + i \begin{bmatrix} -2 \\ 0 \\ i \end{bmatrix}$$

Finally, plug into the general solution:

$$c_1 e^t \cos \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} - c_2 e^t \sin \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} + c_2 e^t \cos \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + c_4 e^t \sin \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} + c_5 e^t \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

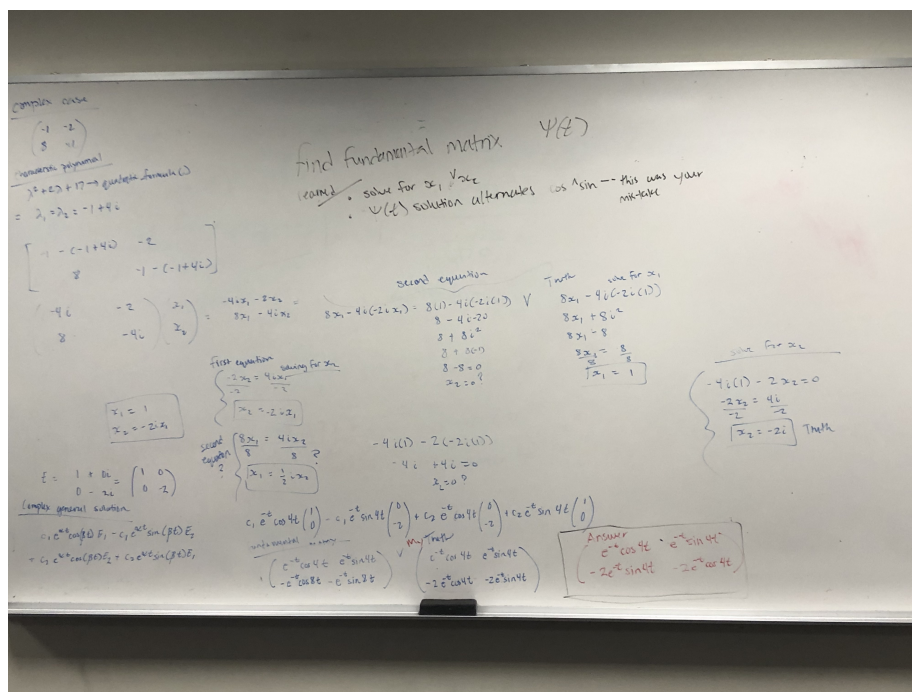
Correct answer:

$$c_1 e^t \cos(t) \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} - c_1 e^t \sin(t) \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} + c_2 e^t \sin(t) \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + c_2 e^t \cos(t) \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} + c_3 e^t \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

5. Find a fundamental matrix for the system $\mathbf{x}(t)' = \mathbf{A}\mathbf{x}(t)$

$$\mathbf{A} = \begin{bmatrix} -1 & -2 \\ 8 & -1 \end{bmatrix}$$

- Almost got this right, need to redo back substitution and build it correctly. Found correct eigen values.
- Note to self: Eigenvalues must be put into the $\cos, \sin$ respective parts for $\alpha \pm \beta$



13. Find the solution to the given system that satisfies the given initial condition

$$\mathbf{x}'(t) = \begin{bmatrix} -3 & -1 \\ 2 & -1 \end{bmatrix} \mathbf{x}(t)$$

$$\mathbf{x}(0) = \begin{bmatrix} -1 \\ 0 \end{bmatrix}, \mathbf{x}(\pi) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \mathbf{x}(-2\pi) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \mathbf{x}(\pi/2) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Algorithm:

1. Find characteristic equation by solving for $\det(\mathbf{A} - \lambda \mathbf{I}) = 0$
2. Check Δ , i.e. check if we are in complex case. Δ if $(b)^2 - 4(a)(c) < 0$

- If in complex case, same steps apply to finding the general solution in the real case, expect we write the general solution as: $c_1 e^{\alpha t} (\cos \beta t E_1 - \sin \beta t E_2) + c_2 e^{\alpha t} (\sin \beta t E_1 + \cos \beta t E_2)$
- Find eigenvectors. For our initial value problem, set up the eigenvalues as the following:

$$\mathbf{x}(0) = \begin{bmatrix} E_{ij} & E_{ij} \\ E_{ij} & E_{ij} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

(do it for each IVP, I showed the set up for $\mathbf{x}(0)$)

- Back substitute the equation in 4, and then write in the following form:

Handwritten work on a whiteboard showing the solution of a system of differential equations. The work includes finding eigenvalues, eigenvectors, and the general solution in both complex and real forms.

Complex case:

Characteristic polynomial: $\lambda^2 + 4\lambda + 17 = 0$ (quadratic formula)

Eigenvalues: $\lambda_1 = -2 + 4i$, $\lambda_2 = -2 - 4i$

Find fundamental matrix $\Psi(t)$

First equation (solving for x_1):

$$\begin{bmatrix} -4 & -2 \\ 8 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -4x_1 - 2x_2 \\ 8x_1 - 4x_2 \end{bmatrix}$$

Second equation (solving for x_2):

$$8x_1 - 4x_2 = 0 \Rightarrow x_2 = 2x_1$$

First equation (solving for x_1):

$$-4x_1 - 2(2x_1) = 0 \Rightarrow -8x_1 = 0 \Rightarrow x_1 = 0$$

Second equation (solving for x_2):

$$8(0) - 4x_2 = 0 \Rightarrow -4x_2 = 0 \Rightarrow x_2 = 0$$

Complex general solution:

$$\mathbf{x}(t) = c_1 e^{(-2+4i)t} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 e^{(-2-4i)t} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Real general solution:

$$\mathbf{x}(t) = e^{-2t} \left(c_1 \begin{bmatrix} \cos 4t \\ \sin 4t \end{bmatrix} + c_2 \begin{bmatrix} -\sin 4t \\ \cos 4t \end{bmatrix} \right)$$

Initial value problem:

$\mathbf{x}(0) = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$

Substitution:

$$\begin{bmatrix} -1 \\ 0 \end{bmatrix} = e^0 \left(c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 2 \end{bmatrix} \right)$$

Solving for c_1 and c_2 :

$$\begin{cases} c_1 - c_2 = -1 \\ 2c_1 + 2c_2 = 0 \end{cases}$$

Final solution:

$$\mathbf{x}(t) = e^{-2t} \begin{bmatrix} -\cos 4t \\ \sin 4t \end{bmatrix}$$

- Note to future me:** Check out the back substitution you did for this problem as it stumped you originally. Also, my work is wrong (starting at the respective eigenspace)

Find the solution, IVP

$$X'(t) = \begin{pmatrix} -3 & -1 \\ 2 & -1 \end{pmatrix} X(t)$$

$$X(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$X(\pi) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$X(-2\pi) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$X(\pi/2) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$\lambda_1 = \lambda_2 = -2 + i$

Characteristic equation

$$\begin{pmatrix} -3-\lambda & -1 \\ 2 & -1-\lambda \end{pmatrix} = 0$$

$$(-3-\lambda)(-1-\lambda) - (-2) = 0$$

$$(3+\lambda)(1+\lambda) + 2 = 0$$

$$\lambda^2 + 4\lambda + 5 = 0$$

If (complex)

$$\text{complex} = (4 \pm 4i) / (2 \pm 2i)$$

$$1 \pm i$$

Then eigenvalue formula

$$X(t) = e^{(1-i)t} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

we see this is the solution

Eigen decomposition

$$\begin{pmatrix} -1-i & -1 \\ 2 & 1-i \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \Rightarrow E = \begin{pmatrix} 1 & 0 \\ -1 & -1 \end{pmatrix}$$

General solution

$$X(t) = e^{(-2+i)t} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + e^{(-2-i)t} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$X(t) = e^{-2t} \left(\cos t \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \sin t \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right)$$

$$X(t) = e^{-2t} \begin{pmatrix} \cos t \\ \sin t \end{pmatrix}$$

Find the solution, IVP

$$X'(t) = \begin{pmatrix} -3 & -1 \\ 2 & -1 \end{pmatrix} X(t)$$

$$X(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$X(\pi) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$X(-2\pi) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$X(\pi/2) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$\lambda_1 = \lambda_2 = -2 + i$

Characteristic equation

$$\begin{pmatrix} -3-\lambda & -1 \\ 2 & -1-\lambda \end{pmatrix} = 0$$

IVP

$$X(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -1 & -1 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

$$\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$X(t) = e^{(-2+i)t} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + e^{(-2-i)t} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$X(t) = e^{-2t} \left(\cos t \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \sin t \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right)$$

$$X(t) = e^{-2t} \begin{pmatrix} \cos t \\ \sin t \end{pmatrix}$$

- Provided $\beta \neq 0$, and after eigenvalue decomposition, $(\forall a, b \in \mathbb{R})[(a, b \neq$

$\{0\} \implies (4)(5)$ are L.I. because each solution can be represented in the eigenspace, where these are real valued and linearly independent.

My answer is too hand wavy. The way to solve this is to show that the determinant of the Wronskian is never 0. Shown below:

Given our conditions, $\mathbf{a}, \mathbf{b}$ are L.I. $\iff \det(W(t)) \neq 0$

Given our preconditions, it suffices to show that from equations (4), (5), are independent $\iff \det(W(t)) \neq 0$, where the Wronskian is made up of $[\mathbf{a} \ \mathbf{b}]$. We can show this by taking any arbitrary point in t on the interval I , and if its $\det(W(t)) \neq 0$ then this implies $\mathbf{a}, \mathbf{b}$ must be linearly independent $\forall t \in \mathbb{R}$, implying $\mathbf{x}_1, \mathbf{x}_2$ are L.I. as well.

$$\mathbf{x}_1(t) = e^{\alpha t}(\cos \beta t \ \mathbf{a} - \sin \beta t \ \mathbf{b})$$

$$\mathbf{x}_2(t) = e^{\alpha t}(\sin \beta t \ \mathbf{a} + \cos \beta t \ \mathbf{b})$$

Lets show it with t_0 :

$$e^0 \cos 0 \ \mathbf{a} - e^0 \sin 0 \ \mathbf{b} = 1 \cdot 1 \cdot \mathbf{a} - 1 \cdot 0 \cdot \mathbf{b} = \mathbf{a}$$

$$e^0 \sin \beta t \ \mathbf{a} + e^0 \cos \beta t \ \mathbf{b} = 1 \cdot 0 \cdot \mathbf{a} + 1 \cdot 1 \cdot \mathbf{b} = \mathbf{b}$$

$$\implies W(t) = [\mathbf{a} \ \mathbf{b}]$$

$$\det(W(t)) \neq 0 \iff [\mathbf{a} \ \mathbf{b}] \text{ L.I. } \forall t \in \mathbb{R}$$

21. RLC Network: The currents in the RLC network given by the schematic diagram in Figure 9.6 are governed by the following equations:

$$4I_2'(t) + 52q_1(t) = 10$$

$$13I_3(t) + 52q_1(t) = 10$$

$$I_1(t) = I_2(t) + I_3(t)$$

where $q_1(t)$ is the change on the capacitor, $I_1(t) = q_1'(t)$, and initially $q_1(0) = 0$ coulombs and $I_1(0) = 0$ amps. Solve for the currents I_1, I_2, I_3 . [Hint: Differentiate the first two equations, eliminate I_1 , and form a normal system with $x_1 = I_2, x_2 = I_2', x_3 = I_3$]

$$4I_2'(t) + 52q_1(t) = 10,$$

$$13I_3(t) + 52q_1(t) = 10,$$

$$I_1(t) = I_2(t) + I_3(t),$$

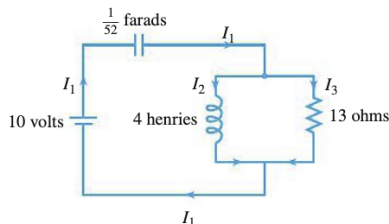


Figure 9.6 RLC network for Problem 21

$$\begin{aligned}
(1) \quad & 4I_2'(t) + 52q_1(t) = 10 \\
(2) \quad & 13I_3(t) + 52q_1(t) = 10 \\
(3) \quad & I_1(t) = I_2(t) + I_3(t)
\end{aligned}$$

Algo:

- a. If we are given $I_1(t) = q^1 t$, then set up equation in normal form.
 - Proceed by eliminating $I_1(t)$ through the branch junction (i.e. (3)) –by letting $x_1 = I_2(t), x_2' = I_2'(t), x_3 = I_3(t)$
 - After differentiation, $(1)' = I_2''(t) = -13x_1 - 13x_3$, $(2)' = -4x_1 + 4x_3$ with $x_1' = x_2$. In normal form that gives:

$$\begin{bmatrix} 0 & 1 & 0 \\ -4 & 0 & -4 \\ -13 & 0 & -13 \end{bmatrix}$$

- b. We evaluate the systems of equations at 0 because our original physical law evaluated eq. holds $\forall t \in \mathbb{R}$. We are given a specific time so we proceed with that, i.e. $q_1(0) = 0$ is given. This gives:

$$q_1(0) = 0 \implies 4I_2'(t) + 52 \cdot 0 = 10 = 10/4$$

$$q_1(0) = 0 \implies 13I_3(t) + 52 \cdot 0 = 10 = 10/13$$

We also show it for $3 = -10/13$. I am not sure why

- With our normal form equation, conduct eigendecomposition. With eigenvector, back substitute. After writing in the general solution take the constant scalars and $c_1, \dots, c_j$ out. Then, set equal to what was found in step above us.

10.3

Useful Resources

- 3Blue1Brown on Fourier
- 2016 Math 54 Course Homework
- Better explanation than fuck ass textbook
- Probably everything you need to do Fourier

5. Determine whether the given function is even, odd, or neither

$$f(x) = e^{-x} \cos 3x$$

$$f(x) = e^{-x} \cos 3x$$

$$-f(x) = \text{odd} \iff -f(x) = f(-x)$$

$$e^{-(-x)} \cos 3(-x) = e^x \cos -3x \quad \text{cos forces the func to be even so } = e^x \cos 3x \neq -e^x \cos 3x$$

now we show it for even case:

$$\text{even} \iff -x \in f(x) = f(x)$$

$$e^{-(-x)} \cos(3 - (x)) = e^{-x} \cos 3x \neq e^x \cos 3x$$

Thus it is neither

7. Prove the following:

- a. If f, g are even functions, then so is the product fg
- b. If f, g are odd functions, then fg is an even function.
- c. If f is an even function and g is an odd function, then fg is an odd function

$$\text{a. } f, g := \text{even} \implies fg(x) = \text{even}$$

$$f(x) \cdot g(x) = fg(x) \implies \text{still even}$$

$$\text{b. } f, g := \text{odd} \implies -f(x) = f(-x) \wedge -g(x) = g(-x)$$

$$f(-x) \cdot g(-x) = fg(-x) \rightarrow \text{test evenness}$$

$$-x \in fg(-x) = fg(-(-x)) = fg(x) \implies \text{even function}$$

$$\text{c. } f(-x) = f(x) \wedge -g(x) = g(-x)$$

$$f(-x) \cdot g(x) = fg(-(x)) = fg(-x) = -(fg(x))$$

For the following (9 & 15), compute Fourier series for the given function f on the specified interval. Use a computer or graphing calculator to plot a few partial sums of the Fourier series.

Algo

- Determine parity, it dictates the below odd even cases.
 - If odd, $\sin(x)$ fourier term
 - If even $\cos(x)$ fourier term
 - $\sum_{n=1}^{\infty} \frac{(-1)^n}{\text{something}}$ is a form I noticed that we write it in, at the last step
9. $f(x) = x, -\pi < x < \pi$
 15. $f(x) = e^x, -\pi < x < \pi$
- [Ms Shaw](#)
 - [blackredpen](#)
 - [Example 12, Slide 41/75 Solved with complex](#)

We are tasked with finding the coefficients in $e^x = a_0 + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx))$ 17. Determine the function to which the Fourier series for $f(x)$, given in the indicted problem, converges in Problem 9.

26. Show that the set of functions

$$\left\{ \cos \frac{\pi}{2}x, \sin \frac{\pi}{2}x, \cos \frac{3\pi}{2}x, \sin \frac{3\pi}{2}x, \dots, \cos \frac{(2n-1)\pi}{2}x, \sin \frac{(2n-1)\pi}{2}x, \dots \right\}$$

is an orthonormal system on $[-1, 1]$ with respect to the weight function $w(x) \equiv 1$

- Symmetrical interval, with the integration of an even function $\times$ an odd function, ask you to take the integral of an odd function over symmetric I, implying that it must go to 0.
- Use the definitions of $f(x)$ even: $f(-x) = f(x)$
- Use the definition of $g(-x) := -g(x)$

27. Find the orthogonal expansion (generalized Fourier series) for

$$f(x) = \begin{cases} 0, & -1 < x < 0, \\ 1, & 0 < x < 1, \end{cases}$$

in terms of the orthonormal system of Problem 26.

33. Let $\{f_n(x)\}$ be an orthogonal set of functions on the interval $[a, b]$ w.r.t. the weight function $w(x)$. Show that they satisfy the Pythagorean property

$$\|f_m + f_n\|^2 = \|f_m\|^2 + \|f_n\|^2$$

if $m \neq n$.

Mini Lec by GSI on this chapters homework:

- cont func on $-L$ to $3L$, but we just study it on $-L, L$

- How heat is transferred gives us a func that we can test
- Why we use $[-\pi, \pi]$: cosine has period 2π , stretching to L because of $\cos(\frac{x\pi}{L})$ creates ugly arithmetic. We can work with constants with π
- We always have $\int_{-\pi}^{\pi} \cos(nx)\sin(mx)dx = 0$
- We are given a set of three cases for the inner product for our integrals. Highlighted in the circle
- We have a basis, which is cos and sin
- These functions are periodic, meaning repeating.
- How do you find a_1 ?
- Stone-Weierstrass theorem for proving density (very involved to show), lets us represent $f(x)$ as a sum, Or "an infinite length vector"

10.4

5. Compute the Fourier sine series for the given function

$$f(x) = -1, 0 < x < 1$$

7. Compute the Fourier sine series for the given function

$$f(x) = x^2, 0 < x < \pi$$

9. Compute the Fourier sine series for the given function

$$f(x) = \pi - x, 0 < x < \pi$$

13. Compute the Fourier cosine series for the given function

$$f(x) = e^x, 0 < x < 1$$

15. Compute the Fourier cosine series for the given function

$$f(x) = \sin x, 0 < x < \pi$$

1 Sources

- [Playlist lectures](#)
- [Differential Equations you will see and developing your hueristics](#)
- [Angle Sum and difference identities](#)
- [More blogs](#)

- [Differential Equations solved](#)
- [SHIELDS](#)
- [Prof Notes on Ch 9.4](#)
- [Solving Differential equations examples](#)
- [Brian Sun Notes](#)
- [Pauls online notes on Differential equations](#)
- [Superposition Principle and Undetermined Coefficients revisited Lecture](#)
- [Differential equations calculator](#)
- [Find a General Solution to a Nonhomogeneous DE Using Undetermined Coefficients \(Linear\)](#)
- [Why is \$y = y_h + y_p\$](#)
- [A good textbook describing \$y = y_h + y_p\$](#)
- [What is a state vector and how does it reduce higher order differential equation as a matrix system in normal form](#)
- [Explanation of null space](#)
- [Supplemental textbook probably worth reading](#)
- **Gilbert Strang Da Goat IMO for explaining differential equations**
 - [Lecture](#)
 - [Chapter](#)
- [Good Blog on differential equations](#)